A Delta-Hedged Implied Volatility Mispricing Strategy Based on WLS Calibration of a Quadratic Model

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Abstract

This report proposes a systematic trading framework that detects implied-volatility (IV) mispricing using an weighted least squares (WLS) calibration of a parsimonious quadratic "wing" model for the IV smile. The methodology proceeds in three stages: (i) constructing an out-of-the-money (OTM) IV dataset for each trading date and expiration, (ii) calibrating a theoretical IV curve and detecting statistically significant deviations, and (iii) forming a delta-hedged long/short option strategy designed to profit from mean reversion of IV relative to the calibrated curve.

1 Notation and Data Structure

On each trading date t, for a fixed expiration e (time-to-maturity $T_{t,e}$), let:

- S_t be the underlying spot price,
- K_i be the strike of option i,
- $\sigma_{t,e}^{\text{mkt}}(K_i)$ be the market implied volatility (IV),
- $C_{t,e}^{\text{mkt}}(K_i)$ and $P_{t,e}^{\text{mkt}}(K_i)$ be the market call/put prices (e.g. last trade),
- $v_{t,e}^{C}(K_i)$ and $v_{t,e}^{P}(K_i)$ be call/put trading volumes ,
- r_t be the risk-free rate for the relevant maturity.

We use two datasets. A *signal universe* is constructed for calibration and signal generation by restricting strikes to a moneyness band:

$$\mathcal{U}_{t,e}^{\text{sig}} = \left\{K_i : \; K_i \in \left[0.9\, S_t,\; 1.1\, S_t\right]\right\}. \tag{1}$$

For trade execution, valuation, and position unwinding, we additionally maintain an *execution universe* $\mathcal{U}_{t,e}^{\text{all}}$ that retains all strikes with valid quotes (no moneyness truncation) to ensure positions can always be closed when exit conditions are met.

2 OTM IV Construction

To mitigate put–call inconsistencies and focus on the most informative quotes, we form an OTM IV series by selecting:

$$\sigma_{t,e}^{\text{OTM}}(K_i) = \begin{cases} \sigma_{t,e}^{P}(K_i), & K_i \le S_t, \\ \sigma_{t,e}^{C}(K_i), & K_i > S_t, \end{cases} \tag{2}$$

where σ^P and σ^C denote put and call implied volatilities, respectively. After filtering, we obtain $n_{t,e}$ OTM observations for each (t, e).

3 Quadratic Wing Model and WLS Calibration

3.1 Model Specification

For each date t and expiration e, we posit a quadratic model for the theoretical IV smile:

$$\sigma_{t,e}^{\text{model}}(K) = a_{0,t,e} + a_{1,t,e}K + a_{2,t,e}K^2. \tag{3}$$

Let $x_{t,e} = (a_{0,t,e}, a_{1,t,e}, a_{2,t,e})^\top$.

3.2 Volume-weighted least squares (WLS) objective

Define residuals:

$$\varepsilon_i(x_{t,e}) = \sigma_{t,e}^{\text{OTM}}(K_i) - \sigma_{t,e}^{\text{model}}(K_i). \tag{4}$$

We calibrate the quadratic wing model using volume-weighted least squares:

$$\widehat{x}_{t,e}^{\text{WLS}} = \arg\min_x \sum_{i=1}^{n_{t,e}} w_i\, \varepsilon_i(x)^2, \qquad w_i = \frac{\max(v_i, 0)}{\sum_j \max(v_j, 0)}. \tag{5}$$

3.3 Matrix form

Let $b \in \mathbb{R}^n$ be the vector of observed OTM IV and $A \in \mathbb{R}^{n\times 3}$ be the design matrix. Let $W = \operatorname{diag}(w_1, \ldots, w_n)$. Then

$$\widehat{x}_{t,e}^{\text{WLS}} = (A^\top W A)^{-1} A^\top W b, \tag{6}$$

and OLS is recovered as the special case $W = I$.

4 Mispricing Metrics and Outlier Detection

4.1 Raw Mispricing (Residual)

For each strike:

$$e_{t,e}(K_i) = \sigma_{t,e}^{\text{OTM}}(K_i) - \widehat{\sigma}_{t,e}^{\text{model}}(K_i). \tag{7}$$

Positive $e_{t,e}(K_i)$ indicates IV is above the theoretical curve ("overpriced vol"), while negative indicates IV is below the curve ("underpriced vol").

4.2 Robust standardized mispricing (MAD-based z-score)

We compute the raw mispricing residual:

$$e_{t,e}(K_i) = \sigma_{t,e}^{\text{OTM}}(K_i) - \widehat{\sigma}_{t,e}^{\text{model}}(K_i). \tag{8}$$

To reduce sensitivity to outliers and non-Gaussian residuals, we estimate the residual scale using the median absolute deviation (MAD):

$$\widehat{\sigma}_{t,e}^{\text{rob}} = 1.4826 \cdot \text{median}_i\Big(\big|e_{t,e}(K_i) - \text{median}_j(e_{t,e}(K_j))\big|\Big). \tag{9}$$

The standardized mispricing score is then defined as:

$$z_{t,e}(K_i) = \frac{e_{t,e}(K_i)}{\widehat{\sigma}_{t,e}^{\text{rob}}}. \tag{10}$$

We open trades when $|z_{t,e}(K_i)| \geq z_{\text{enter}}$ and close when $|z_{t,e}(K_i)| \leq z_{\text{exit}}$.

4.3 Standardized Mispricing (Z-score)

For robust and scale-free thresholds, define:

$$z_{t,e}(K_i) = \frac{e_{t,e}(K_i) - \widehat{\mu}_{t,e}}{\widehat{\sigma}_{t,e}}, \tag{11}$$

and select trades when $|z_{t,e}(K_i)|$ exceeds an entry threshold z_{enter}.

5 Trading Strategy Construction

5.1 Signal Generation

On each date t and expiration e, after calibration:

- **Underpriced IV (Buy option):** if $z_{t,e}(K_i) \leq -z_{\text{enter}}$, go long option i.
- **Overpriced IV (Sell option):** if $z_{t,e}(K_i) \geq +z_{\text{enter}}$, go short option i.

In the simplest implementation, the strategy trades OTM options only; in an extended implementation, it may also trade the full filtered set but uses OTM IV for calibration.

5.2 Positioning and Portfolio Interpretation

Let $q_t(i)$ denote the number of option contracts held for instrument i at date t (positive for long, negative for short). The strategy seeks to monetize mean reversion in IV relative to the fitted curve while controlling directional exposure via delta hedging.

5.3 Daily Delta Hedging

Let $\Delta_t(i)$ be the option delta (computed via Black–Scholes using market IV):

$$\Delta_t^C(K) = \Phi(d_1), \qquad \Delta_t^P(K) = \Phi(d_1) - 1, \tag{12}$$

where

$$d_1 = \frac{\ln(S_t/K) + (r_t + \frac{1}{2}\sigma_t^2)T_{t,e}}{\sigma_t\sqrt{T_{t,e}}}, \tag{13}$$

and $\Phi(\cdot)$ is the standard normal CDF. With contract multiplier M (e.g., $M = 100$), the portfolio delta in shares is:

$$\Delta_t^{\text{port}} = \sum_{i \in \mathcal{P}_t} q_t(i)\, M\, \Delta_t(i). \tag{14}$$

A delta-neutral hedge holds underlying shares h_t such that:

$$h_t = -\Delta_t^{\text{port}}. \tag{15}$$

In our implementation, the hedge is rebalanced *daily* to maintain delta neutrality.

5.4 Exit Rules

Positions are closed when any of the following conditions holds:

1. **Mean reversion:** $|z_{t,e}(K_i)| \leq z_{\text{exit}}$ (with $z_{\text{exit}} < z_{\text{enter}}$).
2. **Time stop:** holding period exceeds H trading days.
3. **Maturity stop:** remaining DTE falls below a minimum threshold .

5.5 Transaction Cost Model (Entry/Exit + Daily Hedging)

To keep the backtest tractable while accounting for realistic implementation frictions, we apply a transaction cost model that charges costs (i) when opening/closing option positions and (ii) when rebalancing the daily delta hedge in the underlying.

Option trading costs (charged at entry and exit). For an option trade of $\Delta q_t(i)$ contracts executed at the price proxy $p_t(i)$ with contract multiplier M, we charge a fixed commission per contract per side:

$$\mathrm{TC}_t^{\mathrm{opt}}(i) = |\Delta q_t(i)|\, c_{\mathrm{opt}}, \tag{16}$$

where c_{opt} is the per-contract commission per side. Accordingly, the total option cost for a position equals the sum of costs at entry and at exit.

Underlying hedging costs (charged at each hedge rebalance). Let h_t denote the number of underlying shares held for delta hedging at date t, and $\Delta h_t = h_t - h_{t-1}$ be the hedge adjustment (shares traded) executed at the underlying price S_t. In our implementation we set the per-share fee to zero, and model hedging costs using a proportional slippage in basis points:

$$\mathrm{TC}_t^{\mathrm{und}} = |\Delta h_t|\, S_t \left(\frac{b}{10{,}000}\right), \tag{17}$$

where b is the slippage in basis points (bps).

Total transaction costs. The total transaction cost over the backtest horizon is the sum of all option entry/exit costs and all daily hedge rebalancing costs:

$$\mathrm{TC}^{\mathrm{total}} = \sum_t \sum_i \mathrm{TC}_t^{\mathrm{opt}}(i) + \sum_t \mathrm{TC}_t^{\mathrm{und}}. \tag{18}$$

6 Backtesting Protocol

A daily backtest loop proceeds as follows for each trading date t:

1. **Data preprocessing:** filter strikes by moneyness band, remove invalid IV/price, enforce liquidity constraints.
2. **Calibration:** for each expiration e, fit $\widehat{\sigma}_{t,e}^{\mathrm{model}}(K)$ using WLS.
3. **Mispricing detection:** compute $e_{t,e}(K_i)$ and $z_{t,e}(K_i)$; flag candidates with $|z| \geq z_{\mathrm{enter}}$.
4. **Trade execution:** open long (underpriced IV) and short (overpriced IV) positions subject to position limits; deduct transaction costs.
5. **Hedging:** compute Δ_t^{port} and rebalance underlying hedge to target delta neutrality; deduct hedging costs.
6. **Exit management:** close positions that satisfy exit rules; deduct transaction costs.
7. **Performance accounting:** compute daily P&L and equity curve.

7 Trading Strategy Backtesting

7.1 Empirical Backtest Parameters

Table 1 summarizes the empirical parameter settings used in the backtest implementation.

Table 1: Backtest parameter settings.

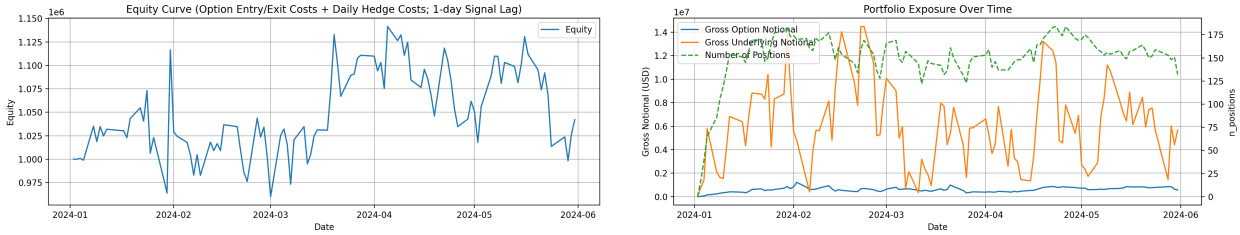
Parameter	**Symbol**	**Value**	**Description**
Entry threshold (z-score)	z_{enter}	2.0	Open trades if $z \leq -z_{\text{enter}}$ (buy) or $z \geq z_{\text{enter}}$ (sell).
Exit threshold (z-score)	z_{exit}	0.5	Close trades when $\lvert z \rvert \leq z_{\text{exit}}$ (mean-reversion exit).
Max trades per expiry per day	N_{tr}	1	For each expiration and day, select up to one most underpriced and one most overpriced OTM option.
Max holding days (time stop)	H	10	Force exit if a position is held longer than H trading days.
Min DTE for trading/holding	DTE_{min}	3	Force exit if remaining DTE is below DTE_{min}.
Contracts per trade	q_0	1	Trade size per signal (contracts).
Minimum volume filter	V_{min}	1	Exclude options with volume $< V_{\text{min}}$.
Minimum option price filter	p_{min}	0.01	Exclude options with price proxy $< p_{\text{min}}$.
Initial cash	E_0	1,000,000	Initial portfolio value in USD.
Option multiplier	M	100	Contract multiplier (e.g., 100 shares per option contract).
Option commission (per contract, per side)	c_{opt}	0.50	Fixed option commission used in TC^{opt}.
Underlying slippage (bps)	b	1.0	Underlying hedge slippage in basis points in TC^{und}.

7.2 Backtest Outputs

Table 2: Backtest performance summary.

Metric	**Value**
Start Equity	1,000,000.00
End Equity	1,042,027.99
Total Return	4.20%
Annualized Sharpe Ratio	0.438
Maximum Drawdown (MDD)	14.02% (Peak: 2024-01-31; Trough: 2024-03-01)

Figures 1–3 present the key backtest outputs: the equity curve, exposure dynamics, cumulative returns with maximum drawdown (MDD) annotation, a tabular snapshot of the backtest log, and an illustrative mispricing detection plot with call/put labels.



(a) Equity curve.

(b) Gross notionals and number of positions.

Figure 1: Backtest equity and exposure diagnostics.

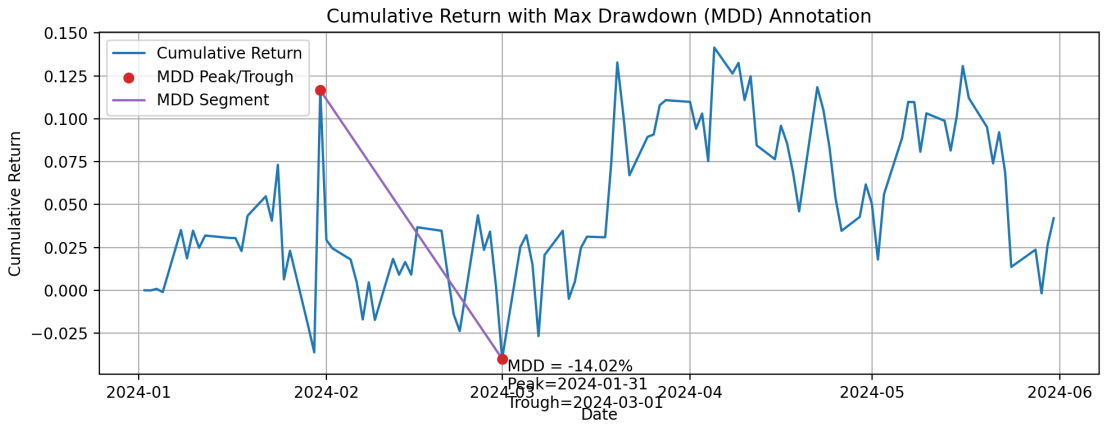


Figure 2: Cumulative return with maximum drawdown (MDD) annotation.

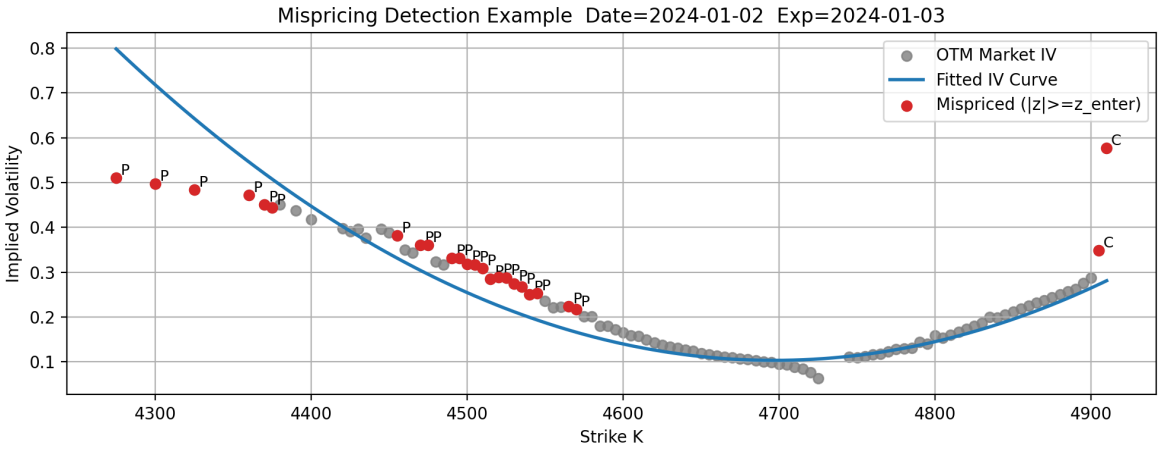


Figure 3: Mispricing detection example for a given date and expiration. The fitted IV curve is shown with flagged mispriced points ($|z| \geq z_{\text{enter}}$). Each flagged point is annotated by `C` or `P` to indicate whether the OTM IV is taken from the call or put side.

Backtest Log (Head 20 + Tail 20)

date	equity	cash	option_value	underlying_value	hedge_shares	n_positions
2024-01-02	1,000,000.00	1,000,000.00	0.00	0.00	0.00	0
2024-01-03	999,915.75	1,746,837.80	10,533.00	-757,455.05	-161.00	17
2024-01-04	1,000,814.32	-436,635.62	24,513.00	1,412,936.93	301.35	39
2024-01-05	998,952.64	-4,785,178.19	-8,941.00	5,793,071.83	1,233.29	65
2024-01-08	1,035,115.04	-1,038,660.85	5,569.00	2,068,206.89	434.17	86
2024-01-09	1,018,658.37	2,656,891.21	7,306.00	-1,645,538.84	-345.96	106
2024-01-10	1,034,752.58	2,573,429.88	4,072.00	-1,542,749.30	-322.52	119
2024-01-11	1,024,796.69	5,057,910.40	20,221.00	-4,053,334.71	-847.94	136
2024-01-12	1,031,914.13	7,826,026.12	33,318.00	-6,827,429.99	-1,427.19	154
2024-01-16	1,030,538.33	7,406,254.13	-1,448.00	-6,374,267.80	-1,337.45	151
2024-01-17	1,030,431.48	5,381,253.45	-21,848.00	-4,328,973.97	-913.44	145
2024-01-18	1,022,878.73	7,843,015.80	37,000.00	-6,857,137.06	-1,434.27	158
2024-01-19	1,043,438.65	9,721,527.48	145,526.00	-8,823,614.83	-1,823.13	169
2024-01-22	1,054,789.38	9,606,981.82	166,730.00	-8,718,922.44	-1,797.56	170
2024-01-23	1,040,563.09	9,127,008.79	212,358.00	-8,298,803.69	-1,705.96	146
2024-01-24	1,073,155.16	11,210,180.26	262,496.00	-10,399,521.10	-2,136.06	150
2024-01-25	1,006,377.55	5,064,462.39	172,675.00	-4,230,759.84	-864.45	165
2024-01-26	1,023,063.49	9,130,570.63	206,580.00	-8,314,087.14	-1,699.89	177
2024-01-29	979,196.88	9,533,239.18	176,430.00	-8,730,472.30	-1,771.63	179
2024-01-30	963,849.83	14,434,624.36	299,105.00	-13,769,879.53	-2,795.93	182
...						
2024-05-03	1,055,993.29	-581,847.89	-90,279.00	1,728,120.18	337.01	172
2024-05-06	1,088,954.67	3,989,789.59	-22,505.00	-2,878,329.92	-555.58	159
2024-05-07	1,109,827.21	7,871,711.72	51,741.00	-6,813,625.51	-1,313.42	157
2024-05-08	1,109,710.97	9,387,324.19	71,047.00	-8,348,660.23	-1,609.33	153
2024-05-09	1,080,788.77	12,223,492.88	64,445.00	-11,207,149.11	-2,149.40	156
2024-05-10	1,103,177.36	11,671,538.97	103,717.00	-10,672,078.61	-2,043.41	154
2024-05-13	1,098,825.16	9,125,413.07	53,649.00	-8,080,236.91	-1,547.52	158
2024-05-14	1,081,497.23	8,140,507.21	57,841.00	-7,116,850.98	-1,356.45	152
2024-05-15	1,101,369.15	7,401,418.04	139,373.00	-6,439,421.89	-1,213.12	149
2024-05-16	1,130,745.95	9,803,734.70	222,299.00	-8,895,287.75	-1,679.27	157
2024-05-17	1,112,089.23	7,049,510.87	190,943.00	-6,128,364.64	-1,155.58	158
2024-05-20	1,095,161.38	9,269,080.00	278,944.00	-8,452,862.62	-1,592.44	164
2024-05-21	1,074,010.78	6,751,889.66	250,049.00	-5,927,927.88	-1,113.98	158
2024-05-22	1,092,110.49	8,266,828.25	231,951.00	-7,406,668.76	-1,395.64	150
2024-05-23	1,068,368.98	8,401,436.02	215,047.00	-7,548,114.04	-1,432.87	152
2024-05-24	1,013,631.76	6,483,607.81	185,393.00	-5,655,369.06	-1,066.10	159
2024-05-28	1,023,687.87	-372,198.32	-60,594.00	1,456,480.19	274.49	153
2024-05-29	998,259.66	-4,799,949.85	-214,619.00	6,012,828.52	1,141.61	148
2024-05-30	1,026,488.91	-3,342,665.77	-58,758.00	4,427,912.68	845.75	151
2024-05-31	1,042,027.99	6,548,995.36	156,967.00	-5,663,934.37	-1,073.22	132

Figure 4: Backtest.

8 Discussion

The backtest yields a total return of 4.20%, while the annualized Sharpe ratio is 0.438, indicating only moderate risk-adjusted performance. A primary reason is that some detected "mispricings" do not mean-revert to the fitted IV curve within the holding window. In particular, the quadratic smile is an approximation and may label persistent skew or volatility risk premia as mispricing, so the deviation can remain. Moreover, positions are sometimes forced to close due to the time-stop and minimum-DTE constraints before reversion occurs, which weakens the average edge. Finally, daily delta hedging increases turnover and trading frictions, raising return volatility and further lowering the Sharpe ratio.